ONE-PAGE BRIEF

Post-quantum cryptography review: fireblocks-mpc-lib @ 00ae08b7f1bd

Client: Fireblocks — public repository (passive review) · Source: github.com/fireblocks/mpc-lib, commit `00ae08b7f1bd` · Reviewed: 2026-11-04 18:44 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **0** High **7** Medium
0 Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium	QSI-2026-11900	**Quantum-vulnerable Paillier: threshold signing and MPC protocols (30 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium	QSI-2026-11901	**Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (8 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium	QSI-2026-11902	**Quantum-vulnerable Threshold ECDSA: threshold signing and MPC protocols (7 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium	QSI-2026-11903	**Quantum-vulnerable FROST: threshold signing and MPC protocols (7 file(s))** confirmed by the analyst What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...

3 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium threshold signing and MPC protocols	Paillier, Pedersen commitment, FROST, Threshold ECDSA ...	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the...
Medium role not determined	Ed25519	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Analyst review: 7 confirmed, 0 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-28)

Static review of source code and manifests; files analysed: 123. No requests were sent to the owner's systems.

Full report, content SHA-256: `6b82d51a20d754131c396986636182db84f3dd9fa04205edb074c580b3336b59`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.